

Assignment #2 Self-Evaluation

Roll-number: _____

Section: _____

Category	Criteria	Marks	Obtained
Problem Representation & Initialization	Chromosome representation: list of 20 genes (P1 to P20), each a valid zone Z1 to Z8	5	
	Initial population generation: configurable size, random valid chromosomes produced	5	
Genetic Algorithm Components	Selection strategy: one of roulette wheel, tournament, or rank selection correctly implemented	8	
	Crossover: single-point or two-point crossover producing valid offspring	8	
	Mutation: random zone reassignment per gene with configurable probability	5	
	Termination condition: stops at max generations or when a satisfactory fitness level is reached	4	
Fitness Function & Constraint Handling	Correct implementation of all 9 constraints (zone capacity, fragile, hazardous, temp, heavy, Z6, Z8, compatibility, food-chemical)	15	
	Accurate penalty calculations: correct penalty value applied per violation type	5	
	Conflict and edge case handling: products with multiple constraints handled consistently and clearly explained in implementation	5	
	Logical correctness: fitness function returns lower score for better solutions	5	
Algorithm Performance & Convergence	Improvement over generations: fitness score visibly decreases across generations	5	
	Reasonable final fitness: solution achieves a reasonably low penalty score	5	
	Proper parameter usage: population size and generation count are configurable and justified	5	
Output & Results	Correct storage plan output: products listed per zone in required format	4	
	Fitness score displayed: best fitness score printed at end of run	2	
	Generation count displayed: number of generations printed	2	
	Violations listed: any remaining constraint violations reported	2	
Report / Documentation	Explanation of GA design choices: chromosome, selection, crossover and mutation described	4	
	Justification of parameters: population size and generation count reasoned	3	
	Clarity and completeness: report is well-structured and covers all required points	3	
Bonus: Excel Export	Optimised storage plan exported to .xlsx with all required columns and clear formatting	5	
TOTAL		100	